WASTE TO WEALTH

Over time, it fragments into micro and nano-sized particles, which infiltrate the soil, air, water and vegetables, causing health issues in humans and animals (see *Growing Sea of Plastic*). The presence of plastic in complex body parts such as heart and brain suggests it is not only being ingested (reflected by its presence in urine and faeces) but entering the bloodstream in the form of microplastics and accumulating in various organs.

Animals often mistake plastic for food and end up eating it. Dr P.V. Parikh, professor and head of veterinary surgery and radiology at the College of Veterinary Science and Animal Husbandry, Kamdhenu University in Anand, Gujarat, recalls operating on a cow that had 65 kg of plastic bags, cups and disposable spoons entangled in its stomach. The owner had bought the cow believing she was pregnant, but when she did not deliver in nine months, he brought her to Parikh. Garima Poonia, whose solid waste management initiative Kachrewala Foundation has removed over four tonnes of plastic waste from the beaches of the Andaman Islands since 2018, says plastic fishing nets get entangled in corals, eventually suffocating, starving and poisoning them. "Last year alone, we recovered more than a kilometre of fishing line and ropes," she says.

WHAT HAS BEEN DONE

To reduce plastic use, the government in 2022 banned 19 single-use plastic (SUP) items with "low utility and high littering potential", including plastic cutlery, straws, trays, decorative thermocol. Additionally, a thickness criterion was imposed on SUP items like carry-bags and banners—plastic carry bags less than 120 microns and PVC banners less than 100 microns were banned as it is difficult to collect and recycle thin plastics.

Meanwhile, in an effort to check the amount of plastic ending up in landfills, the government notified the Guidelines on Extended Producer Responsibility (EPR) for Plastic Packaging in 2022, wherein producers, importers and brand owners (PIBOs) were obligated for 100 per cent of life disposal of the plastic generated from the packaging of their products. This included targets for collection, recycling, end-of-life management, recycled content usage and selective reuse in plastic packaging. "As on date, it is among the world's biggest frameworks for EPR implementation on plastic packaging," says a senior official of the environment ministry. Beginning this April, beverage companies, including Coke and PepsiCo, are required to use 30 per cent recycled plastic in their bottles. A Hindustan Unilever spokesperson welcomed the mandated use of recycled plastic in packaging material. "From 58,000 TPA in 2020, HUL increased its EPR quantity to over 100,000 TPA in 2021 and has been sustaining at 100 per cent of our plastic footprint since then," he says. Likewise, Tejashree Joshi, head of environmental sustainability, Godrej Enterprises Group, says they have been offsetting 10-15 per cent more than the volume of plastic used for packaging their products through their recycling partners. "Over the past five years, we have recycled more than 15,000 tonnes of plastic from various regions."

A few exceptions apart, good intentions, as with everything else in India, falter at the doorstep of implementation. According to a report by Australian philanthropy Minderoo Foundation, India's SUP ban addressed hardly 11 per cent of the products in the category, and its implementation has been ineffective on many fronts. A survey conducted by Delhi-based non-profit Toxics Link in October 2023 at 700 survey points—including street food vendors, restaurants, railway stations and malls—across five



"PLASTICS POSE A CHALLENGE AT EVERY STAGE OF THEIR LIFE CYCLE—FROM PRODUCTION TO DISPOSAL. TO PREVENT PLASTIC POLLUTION, EACH STAKEHOLDER HAS TO PLAY A PART"

SIDDHARTH GHANSHYAM SINGH
Programme Manager, Solid Municipal Waste, CSE



POLYETHYLENE TEREPHTHALATE (PET)

Used in: Disposable water bottle, polyester cloth, ropes, binding straps, covering seats

Recyclability: Yes, widely recycled



HIGH DENSITY POLYETHYLENE (HDPE)

Used in: Shampoo bottles, jerry cans, chairs, bottle crates, fertiliser bags

Recyclability: Yes, widely recycled